

Petri Nets as Model for Session Typed π -calculus

@ Oldenburg

Dan Frumin

University of Groningen

March 9, 2026



Session Typed π -calculus

Interpreting LL Propositions as Session Types

$!U; S$	<i>send value</i> of type U , continue as S	$U \otimes S$
$?U; S$	<i>receive value</i> of type U , continue as S	$U \multimap S$
end	<i>terminate</i> the session	$\mathbf{1}$
$\&\{l_1 : A; l_2 : B\}$	<i>offer</i> the choice	$A \& B$
$\oplus\{l_1 : A; l_2 : B\}$	<i>select</i> the choice	$A \oplus B$

Propositions as Sessions: $\otimes$

$$\frac{\Delta_1 \vdash P :: y : A \quad \Delta_2 \vdash Q :: x : B}{\Delta_1, \Delta_2 \vdash (\nu y). (\bar{x} \langle y \rangle (P \parallel Q)) :: x : A \otimes B}$$

Propositions as Sessions: $\otimes$

$$\frac{\Delta_1 \vdash P :: y : A \quad \Delta_2 \vdash Q :: x : B}{\Delta_1, \Delta_2 \vdash (\nu y). (\bar{x}\langle y \rangle (P \parallel Q)) :: x : A \otimes B}$$



Send y over
 x

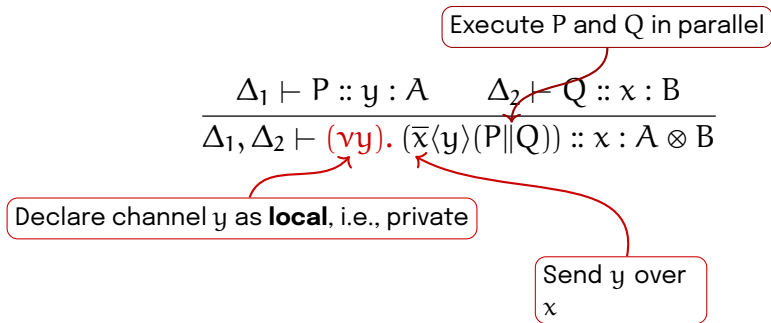
Propositions as Sessions: $\otimes$

$$\frac{\Delta_1 \vdash P :: y : A \quad \Delta_2 \vdash Q :: x : B}{\Delta_1, \Delta_2 \vdash (\nu y). (\bar{x}\langle y \rangle (P \parallel Q)) :: x : A \otimes B}$$

Execute P and Q in parallel

Send y over x

Propositions as Sessions: $\otimes$



Propositions as Sessions: $\otimes$

$$\frac{\Delta_1 \vdash P :: y : A \quad \Delta_2 \vdash Q :: x : B}{\Delta_1, \Delta_2 \vdash (\nu y). (\bar{x} \langle y \rangle (P \parallel Q)) :: x : A \otimes B}$$

$$\frac{\Delta, y : A, x : B \vdash R :: z : C}{\Delta, x : A \otimes B \vdash x(y).R :: z : C}$$

Receive y on x

Propositions as Session: $A \multimap B$

For $A \multimap B$, we have a symmetric situation:

$$\frac{\Delta, y : A \vdash P :: z : B}{\Delta \vdash z(y).P :: z : A \multimap B} \qquad \frac{\Delta_1 \vdash P :: y : A \quad x : B, \Delta_2 \vdash Q :: z : C}{\Delta_1, x : A \multimap B, \Delta_2 \vdash (\nu y). (\bar{x}\langle y \rangle (P \parallel Q)) :: z : C}$$

Propositions as Session: $A \multimap B$

For $A \multimap B$, we have a symmetric situation:

$$\frac{\Delta, y : A \vdash P :: z : B}{\Delta \vdash z(y).P :: z : A \multimap B}$$

$$\frac{\Delta_1 \vdash P :: y : A \quad x : B, \Delta_2 \vdash Q :: z : C}{\Delta_1, x : A \multimap B, \Delta_2 \vdash (\nu y). (\bar{x}\langle y \rangle (P \parallel Q)) :: z : C}$$



To **offer** a ' $\multimap$ ', we implement a receive

Propositions as Session: $A \multimap B$

For $A \multimap B$, we have a symmetric situation:

$$\frac{\Delta, y : A \vdash P :: z : B}{\Delta \vdash z(y).P :: z : A \multimap B}$$

$$\frac{\Delta_1 \vdash P :: y : A \quad x : B, \Delta_2 \vdash Q :: z : C}{\Delta_1, x : A \multimap B, \Delta_2 \vdash (\nu y). (\bar{x}\langle y \rangle (P \parallel Q)) :: z : C}$$

To **rely** on a ' $\multimap$ ', we implement a send

To **offer** a ' $\multimap$ ', we implement a receive

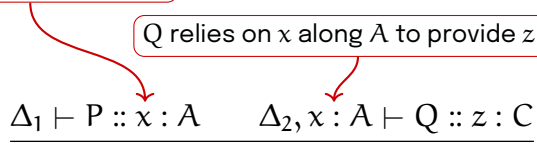
Process composition: cut

$$\frac{\Delta_1 \vdash P :: x : A \quad \Delta_2, x : A \vdash Q :: z : C}{\Delta_1, \Delta_2 \vdash (\nu x).(P \parallel Q) :: z : C}$$

Process composition: cut

P can provide A along x

Q relies on x along A to provide z : C


$$\frac{\Delta_1 \vdash P :: x : A \quad \Delta_2, x : A \vdash Q :: z : C}{\Delta_1, \Delta_2 \vdash (\nu x).(P \parallel Q) :: z : C}$$

Process composition: cut

P can provide A along x

Q relies on x along A to provide z : C

$$\frac{\Delta_1 \vdash P :: x : A \quad \Delta_2, x : A \vdash Q :: z : C}{\Delta_1, \Delta_2 \vdash (\nu x).(P \parallel Q) :: z : C}$$

Execute P and Q in parallel

Declare channel x **local** to P and Q

Process composition: cut

P can provide A along x

Q relies on x along A to provide z : C

$$\frac{\Delta_1 \vdash P :: x : A \quad \Delta_2, x : A \vdash Q :: z : C}{\Delta_1, \Delta_2 \vdash (\nu x).(P \parallel Q) :: z : C}$$

Execute P and Q in parallel

Declare channel x **local** to P and Q

Cut removal in logic is crucial to the dynamics of π MILL.

Principal Cut Reductions: Synchronization via $\otimes$

$$\frac{\frac{\Delta_1 \vdash P :: y : A \quad \Delta_2 \vdash Q :: x : B}{\Delta_1, \Delta_2 \vdash (\nu y).(\bar{x}\langle y \rangle.(P \parallel Q)) :: x : A \otimes B} \quad \frac{\Delta_3, y : A, x : B \vdash R :: z : C}{\Delta_3, x : A \otimes B \vdash x(y).R :: z : C}}{\Delta_1, \Delta_2, \Delta_3 \vdash (\nu x).((\nu y).\bar{x}\langle y \rangle.(P \parallel Q) \parallel x(y).R) :: z : C}$$

$\longrightarrow$

Principal Cut Reductions: Synchronization via $\otimes$

$$\frac{
 \frac{
 \frac{\Delta_1 \vdash P :: y : A}{\Delta_1, \Delta_2 \vdash (\nu y).(\bar{x}\langle y \rangle).(P \parallel Q)) :: x : A \otimes B}
 \quad
 \frac{\Delta_2 \vdash Q :: x : B}{\Delta_3, x : A \otimes B \vdash x(y).R :: z : C}
 }{
 \Delta_1, \Delta_2, \Delta_3 \vdash (\nu x).((\nu y).\bar{x}\langle y \rangle).(P \parallel Q) \parallel x(y).R) :: z : C
 }
 \longrightarrow$$

Principal Cut Reductions: Synchronization via $\otimes$

$$\begin{array}{c}
 \frac{\frac{\Delta_1 \vdash P :: y : A \quad \Delta_2 \vdash Q :: x : B}{\Delta_1, \Delta_2 \vdash (\nu y).(\bar{x}\langle y \rangle.(P \parallel Q)) :: x : A \otimes B} \quad \frac{\Delta_3, y : A, x : B \vdash R :: z : C}{\Delta_3, x : A \otimes B \vdash x(y).R :: z : C}}{\Delta_1, \Delta_2, \Delta_3 \vdash (\nu x).((\nu y).\bar{x}\langle y \rangle.(P \parallel Q) \parallel x(y).R) :: z : C} \\
 \longrightarrow \\
 \frac{\Delta_1 \vdash P :: y : A \quad \Delta_3, y : A, x : B \vdash R :: z : C}{\Delta_1, x : B, \Delta_3 \vdash (\nu y).(P \parallel R) :: z : C}
 \end{array}$$

Principal Cut Reductions: Synchronization via $\otimes$

$$\begin{array}{c}
 \frac{\Delta_1 \vdash P :: y : A \quad \Delta_2 \vdash Q :: x : B}{\Delta_1, \Delta_2 \vdash (\nu y).(\bar{x}\langle y \rangle).(P \parallel Q)) :: x : A \otimes B} \quad \frac{\Delta_3, y : A, x : B \vdash R :: z : C}{\Delta_3, x : A \otimes B \vdash x(y).R :: z : C} \\
 \hline
 \Delta_1, \Delta_2, \Delta_3 \vdash (\nu x).((\nu y).\bar{x}\langle y \rangle).(P \parallel Q) \parallel x(y).R) :: z : C \\
 \\
 \xrightarrow{\quad} \\
 \frac{\Delta_2 \vdash Q :: x : B \quad \frac{\Delta_1 \vdash P :: y : A \quad \Delta_3, y : A, x : B \vdash R :: z : C}{\Delta_1, x : B, \Delta_3 \vdash (\nu y).(P \parallel R) :: z : C}}{\Delta_1, \Delta_2, \Delta_3 \vdash (\nu x).(Q \parallel (\nu y).(P \parallel R)) :: z : C}
 \end{array}$$

This way, we have the following reduction on processes:

$$(\nu x).((\nu y).\bar{x}\langle y \rangle).(P_1 \parallel P_2) \parallel x(z).Q) \longrightarrow (\nu x).(P_2 \parallel (\nu y).(P_1 \parallel Q[y/z]))$$

Principal Cut Reductions: Synchronization via $\multimap$

The case of $\multimap$ is similar. We have the following:

$$\frac{\frac{\Delta_1, y : A \vdash R :: x : B}{\Delta_1 \vdash x(y).R :: x : A \multimap B} \quad \frac{\Delta_2 \vdash P :: y : A \quad \Delta_3, x : B \vdash Q :: z : C}{\Delta_2, \Delta_3, x : A \multimap B \vdash (\nu y).\bar{x}\langle y \rangle.(P \parallel Q) :: z : C}}{\Delta_1, \Delta_2, \Delta_3 \vdash (\nu x).(x(y).R \parallel (\nu y).\bar{x}\langle y \rangle.(P \parallel Q)) :: z : C}$$

which, omitting large bits of the derivation, can be simplified into

$$\Delta_1, \Delta_2, \Delta_3 \vdash (\nu x).((\nu y).(P \parallel R) \parallel Q) :: z : C$$

Logical correspondence

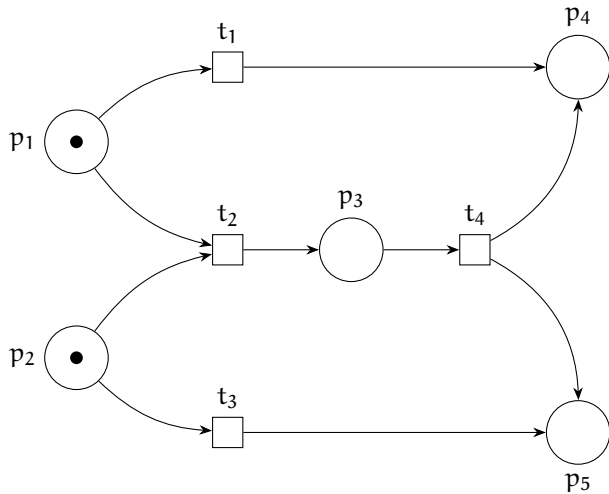
- Reduction on processes correspond to cut reductions in logic, we get deadlock-freedom from it
- Different variations of logic correspond to different type systems
- Studying proof theory of linear logic contributes to study of session type systems

Logical correspondence

- Reduction on processes correspond to cut reductions in logic, we get deadlock-freedom from it
- Different variations of logic correspond to different type systems
- Studying proof theory of linear logic contributes to study of session type systems
- What can we say about the models of linear logic w.r.t. session types?

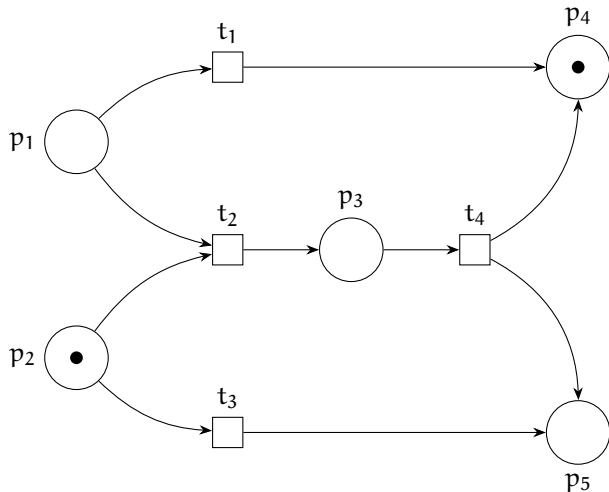
Petri Nets and Linear Logic

Petri Net example



Initial marking: $\{p_1, p_2\}$.

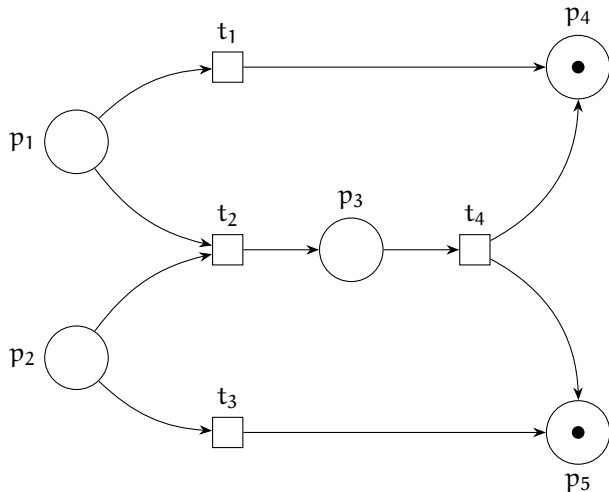
Petri Net example



Initial marking: $\{p_1, p_2\}$.

- Transition t_1 is enabled and can activate, transitioning to $\{p_2, p_4\}$.

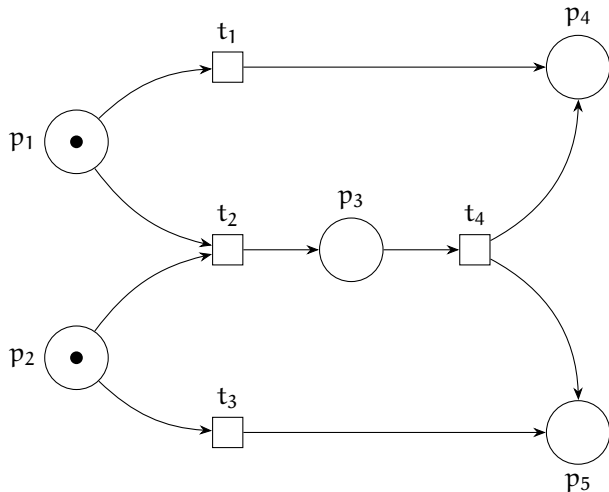
Petri Net example



Initial marking: $\{p_1, p_2\}$.

- Transition t_1 is enabled and can activate, transitioning to $\{p_2, p_4\}$.
- Transition t_3 is enabled and can activate, transitioning to $\{p_4, p_5\}$.
- Note: t_1 and t_3 can activate in any order or in parallel.

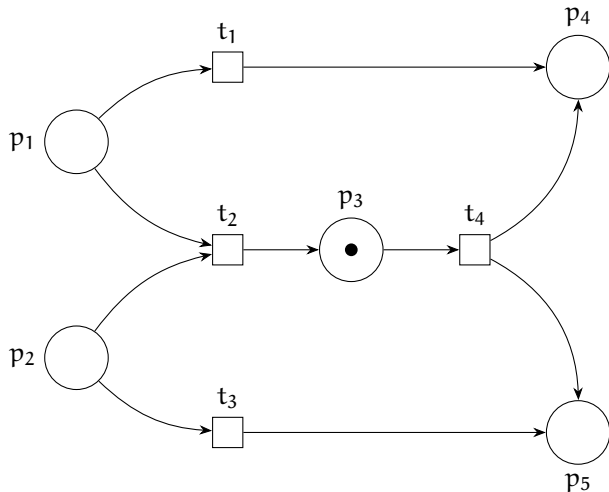
Petri Net example



Initial marking: $\{p_1, p_2\}$.

- Transition t_1 is enabled and can activate, transitioning to $\{p_2, p_4\}$.
- Transition t_3 is enabled and can activate, transitioning to $\{p_4, p_5\}$.
- Note: t_1 and t_3 can activate in any order or in parallel.
- There is another way to reach $\{p_4, p_5\}$ in this net

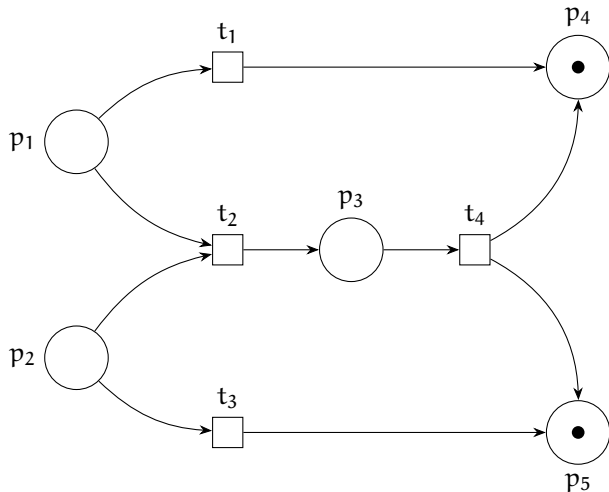
Petri Net example



Initial marking: $\{p_1, p_2\}$.

- Transition t_2 is enabled and can activate, transitioning to p_3 .

Petri Net example



Initial marking: $\{p_1, p_2\}$.

- Transition t_2 is enabled and can activate, transitioning to p_3 .
- Transition t_4 is enabled and can activate, transitioning to $\{p_4, p_5\}$.
- Note: t_4 and t_2 have to fire in sequence.

Petri Nets, formally

A Petri net $N = (P, T, \bullet(-), (-)^\bullet)$ where

$$\bullet(-), (-)^\bullet : T \rightarrow \mathcal{M}(P)$$

describe inputs and outputs of transitions as multisets of places.

Petri Nets, formally

A Petri net $N = (P, T, \bullet(-), (-)\bullet)$ where

$$\bullet(-), (-)\bullet : T \rightarrow \mathcal{M}(P)$$

describe inputs and outputs of transitions as multisets of places.

A **marking** is a multiset M over P . If $\bullet t \subseteq M$, then t is enabled and can activate resulting in the marking $M' = (M \setminus \bullet t) \cup t\bullet$.

Petri nets and $\otimes$ -theories

Petri nets correspond exactly to $\otimes$ -theories [Grunter and Gehlot 1989]:

- Atomic formulas are places $p_i \in P$
- Tensor is the multiset union, 1 is the empty marking
- Each marking $\{p_1, \dots, p_n\}$ corresponds to a formula $p_1 \otimes \dots \otimes p_n$

Petri nets and $\otimes$ -theories

Petri nets correspond exactly to $\otimes$ -theories [Grunter and Gehlot 1989]:

- Atomic formulas are places $p_i \in P$
- Tensor is the multiset union, 1 is the empty marking
- Each marking $\{p_1, \dots, p_n\}$ corresponds to a formula $p_1 \otimes \dots \otimes p_n$
- Transitions $t \in T$ correspond to axioms

$$\bullet t \vdash t \bullet$$

Petri nets and $\otimes$ -theories

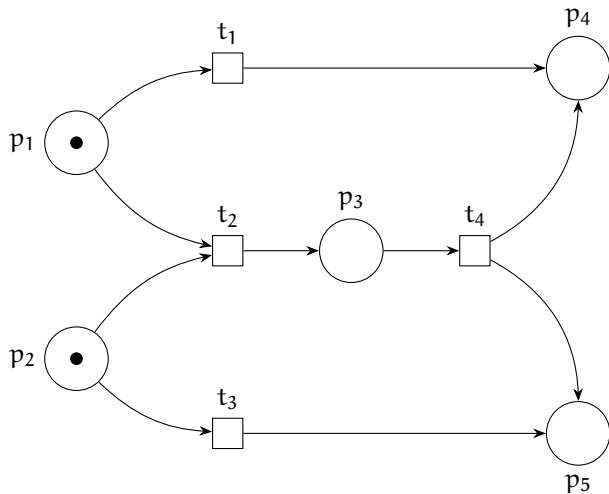
Petri nets correspond exactly to $\otimes$ -theories [Grunter and Gehlot 1989]:

- Atomic formulas are places $p_i \in P$
- Tensor is the multiset union, 1 is the empty marking
- Each marking $\{p_1, \dots, p_n\}$ corresponds to a formula $p_1 \otimes \dots \otimes p_n$
- Transitions $t \in T$ correspond to axioms

$$\bullet t \vdash t \bullet$$

- Derivation $M_1 \vdash M_2$ corresponds to **reachability**

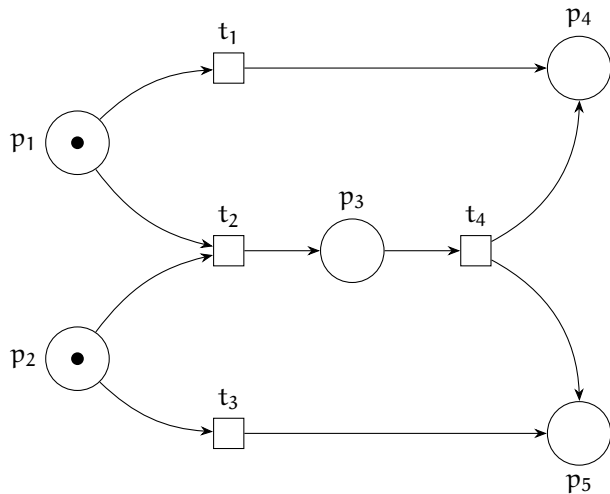
Petri Net example



Derivations in logic:

$$\frac{\frac{p_1 \vdash p_4 \quad p_2 \vdash p_5}{p_1, p_2 \vdash p_4 \otimes p_5}}{p_1 \otimes p_2 \vdash p_4 \otimes p_5}$$

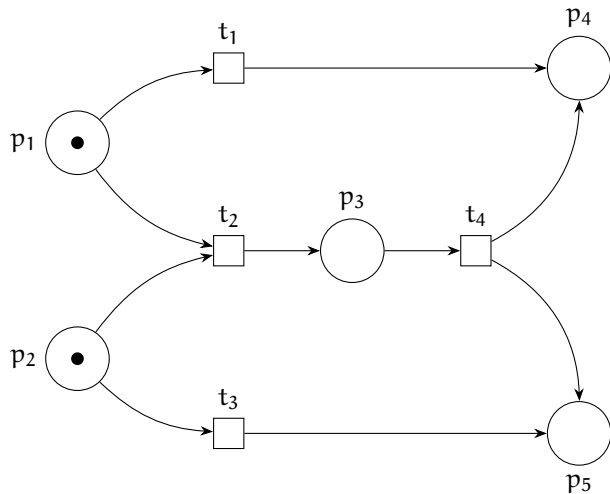
Petri Net example



Transition t_1 transitions in logic: Transition t_3

$$\frac{\frac{p_1 \vdash p_4 \quad p_2 \vdash p_5}{p_1, p_2 \vdash p_4 \otimes p_5}}{p_1 \otimes p_2 \vdash p_4 \otimes p_5}$$

Petri Net example

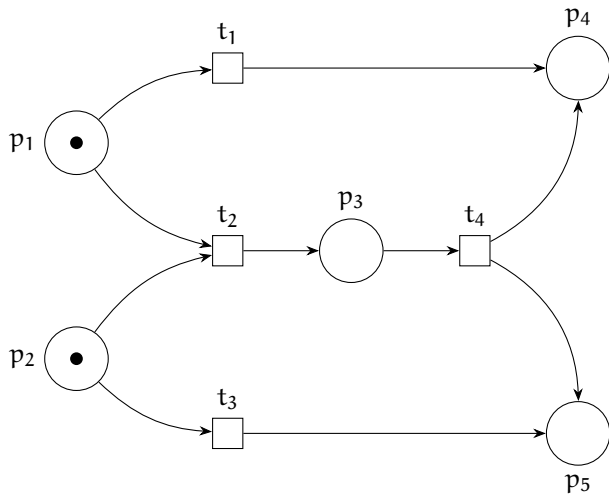


Transition t_1 Transitions in logic: Transition t_3

$$\frac{\frac{p_1 \vdash p_4 \quad p_2 \vdash p_5}{p_1, p_2 \vdash p_4 \otimes p_5}}{p_1 \otimes p_2 \vdash p_4 \otimes p_5}$$

t_1 and t_3 in parallel

Petri Net example

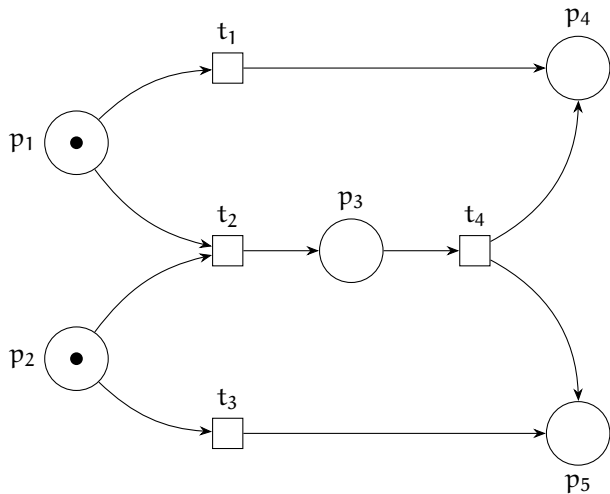


Derivations in logic:

$$\frac{\frac{p_1 \vdash p_4 \quad p_2 \vdash p_5}{p_1, p_2 \vdash p_4 \otimes p_5}}{p_1 \otimes p_2 \vdash p_4 \otimes p_5}$$

$$\frac{p_1 \otimes p_2 \vdash p_3 \quad p_3 \vdash p_4 \otimes p_5}{p_1 \otimes p_2 \vdash p_4 \otimes p_5}$$

Petri Net example



Derivations in logic:

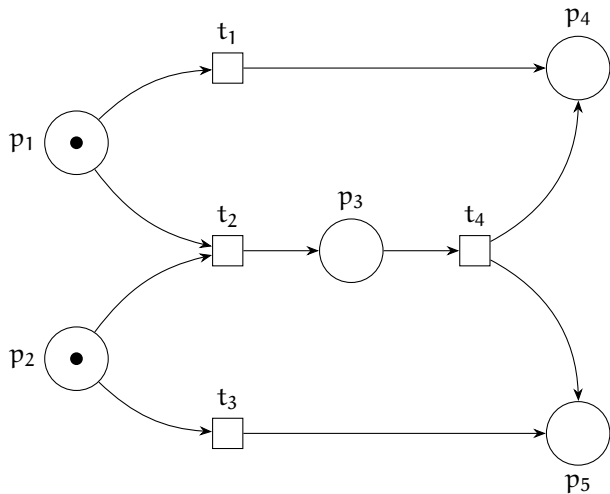
$$\frac{\frac{p_1 \vdash p_4 \quad p_2 \vdash p_5}{p_1, p_2 \vdash p_4 \otimes p_5}}{p_1 \otimes p_2 \vdash p_4 \otimes p_5}$$

$$\frac{p_1 \otimes p_2 \vdash p_3 \quad p_3 \vdash p_4 \otimes p_5}{p_1 \otimes p_2 \vdash p_4 \otimes p_5}$$

Transition t_2

Transition t_4

Petri Net example



Derivations in logic:

$$\frac{\frac{p_1 \vdash p_4 \quad p_2 \vdash p_5}{p_1, p_2 \vdash p_4 \otimes p_5}}{p_1 \otimes p_2 \vdash p_4 \otimes p_5}$$

$$\frac{p_1 \otimes p_2 \vdash p_3 \quad p_3 \vdash p_4 \otimes p_5}{p_1 \otimes p_2 \vdash p_4 \otimes p_5}$$

Transition t_2

Transition t_4

First t_2 then t_4

The category $\mathcal{T}[N]$

Petri nets and $\otimes$ -theories both can be seen as free symmetric monoidal categories. We will need the following: given a Petri net N the category $\mathcal{T}[N]$ is defined in [Meseguer and Montanari 1990] and consists of:

- The objects are markings, i.e. freely generated from $p \in P$.
- The arrows are freely generated from transitions $t : \bullet t \rightarrow t \bullet$.
- We write $t_1 \otimes t_2$ for monoidal product of arrows, and $t_1; t_2$ for composition $t_2 \circ t_1$.

The category $\mathcal{T}[N]$

Petri nets and $\otimes$ -theories both can be seen as free symmetric monoidal categories. We will need the following: given a Petri net N the category $\mathcal{T}[N]$ is defined in [Meseguer and Montanari 1990] and consists of:

- The objects are markings, i.e. freely generated from $p \in P$.
- The arrows are freely generated from transitions $t : \bullet t \rightarrow t \bullet$.
- We write $t_1 \otimes t_2$ for monoidal product of arrows, and $t_1; t_2$ for composition $t_2 \circ t_1$.

Returning to previous example, we have two arrows between $\{p_1, p_2\}$ and $\{p_4, p_5\}$:

- $t_2; t_4$
- $t_1 \otimes t_3 = (t_1 \otimes 1_{p_2}); (1_{p_4} \otimes t_3) = (1_{p_1} \otimes t_3); (t_1 \otimes 1_{p_5})$

Inductive definition of $\mathcal{T}[\mathbb{N}]$

$$\frac{\bullet t = M_1 \quad t^\bullet = M_2}{t : M_1 \rightarrow M_2}$$

$$\frac{M \text{ is a marking}}{\text{id}_M : M \rightarrow M}$$

$$\frac{t : M_1 \rightarrow M_2 \quad u : M_2 \rightarrow M_3}{t; u : M_1 \rightarrow M_3}$$

$$\frac{t : M_1 \rightarrow M_2 \quad u : S_1 \rightarrow S_2}{t \parallel u : M_1 \otimes S_1 \rightarrow M_2 \otimes S_2}$$

$$\frac{t : M_1 \rightarrow M_2 \quad u : M_2 \rightarrow M_3 \quad t' : M'_1 \rightarrow M'_2 \quad u' : M'_2 \rightarrow M'_3}{(t; u) \parallel (t'; u') = (t \parallel t'); (u \parallel u') : M_1 \otimes M'_1 \rightarrow M_3 \otimes M'_3}$$

+ the usual categorical rules

From $\otimes$ -theories to linear logic

So far we know how to interpret $\otimes$ and 1 in a Petri net, where formulas denote markings. How to interpret $\multimap$, $\oplus$, $\&$?

Consider the case for $\multimap$:

$$M_1, M' \vdash M_2 \iff M' \vdash M_1 \multimap M_2,$$

i.e. $M_1 \multimap M_2$ should denote a marking such that if you add M_1 to it you can reach M_2 .

From $\otimes$ -theories to linear logic

So far we know how to interpret $\otimes$ and 1 in a Petri net, where formulas denote markings. How to interpret $\multimap$, $\oplus$, &?

Consider the case for $\multimap$:

$$M_1, M' \vdash M_2 \iff M' \vdash M_1 \multimap M_2,$$

i.e. $M_1 \multimap M_2$ should denote a marking such that if you add M_1 to it you can reach M_2 .
There can be more than 1 (or 0) such markings!

Solution: interpret formulas as **sets of markings**!

Petri nets as models of intuitionistic linear logic

As given in [Engberg and Winskel 1994].

$$\llbracket 1 \rrbracket = \{M \mid \emptyset \text{ is reachable from } M\}$$

$$\llbracket p \rrbracket = \{M \mid \{p\} \text{ is reachable from } M\}$$

$$\llbracket A \otimes B \rrbracket = \{M \mid \exists M_1 \in \llbracket A \rrbracket, M_2 \in \llbracket B \rrbracket, M_1 \cup M_2 \text{ is reachable from } M\}$$

$$\llbracket A \multimap B \rrbracket = \{M \mid \forall M' \in \llbracket A \rrbracket, M \cup M' \in \llbracket B \rrbracket\}$$

$$\llbracket A \& B \rrbracket = \{M \mid M \in \llbracket A \rrbracket \wedge M \in \llbracket B \rrbracket\}$$

$$\llbracket A \oplus B \rrbracket = \{M \mid M \in \llbracket A \rrbracket \vee M \in \llbracket B \rrbracket\}$$

Petri nets as models of intuitionistic linear logic

As given in [Engberg and Winskel 1994].

$$\llbracket 1 \rrbracket = \{M \mid \emptyset \text{ is reachable from } M\}$$

$$\llbracket p \rrbracket = \{M \mid \{p\} \text{ is reachable from } M\}$$

$$\llbracket A \otimes B \rrbracket = \{M \mid \exists M_1 \in \llbracket A \rrbracket, M_2 \in \llbracket B \rrbracket, M_1 \cup M_2 \text{ is reachable from } M\}$$

$$\llbracket A \multimap B \rrbracket = \{M \mid \forall M' \in \llbracket A \rrbracket, M \cup M' \in \llbracket B \rrbracket\}$$

$$\llbracket A \& B \rrbracket = \{M \mid M \in \llbracket A \rrbracket \wedge M \in \llbracket B \rrbracket\}$$

$$\llbracket A \oplus B \rrbracket = \{M \mid M \in \llbracket A \rrbracket \vee M \in \llbracket B \rrbracket\}$$

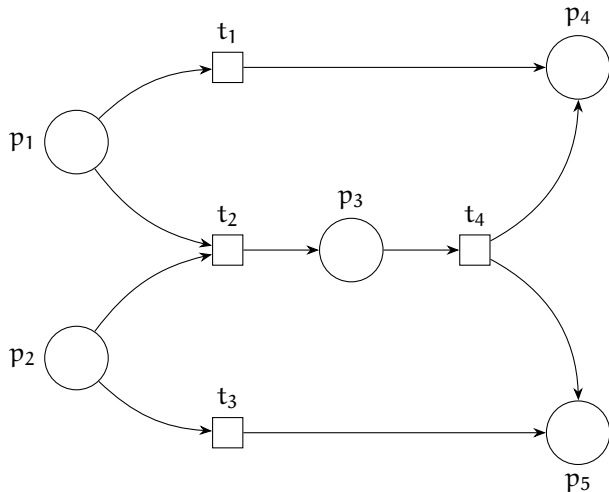
In general: $\llbracket M \rrbracket$ = markings from which M is reachable.

This interpretation is conservative over the $(\otimes, 1)$ -fragment, and:

Lemma

$M \vdash M'$ is derivable iff M' is reachable from M

Expressing properties of Petri nets with MILL



- $p_1 \vdash p_2 \multimap (p_4 \otimes p_5) \oplus p_2$
- $p_1 \vdash p_3 \multimap (p_4 \otimes p_3) \& (p_4 \otimes p_4 \otimes p_5)$
- $p_1 \oplus p_2 \vdash (p_1 \& p_2) \multimap p_3$

Net Interpretation From Linear Logic to Session Types

The category $\widehat{\mathcal{T}}[\mathbf{N}]$

The model of linear logic that we saw concerns with propositions and sets of markings – what happens when we move to the level of types and processes?

The category $\widehat{\mathcal{T}[\mathbf{N}]}$

The model of linear logic that we saw concerns with propositions and sets of markings – what happens when we move to the level of types and processes?

We go from $\mathcal{T}[\mathbf{N}]$ to the presheaves $\widehat{\mathcal{T}[\mathbf{N}]} = \mathcal{T}[\mathbf{N}]^{op} \rightarrow \mathbf{Set}$. The monoidal structure is then lifted:

$$\begin{aligned}1(M) &= \{\mathbf{u} : M \rightarrow \emptyset\} \\(X \otimes Y)(M) &= \{(\mathbf{u}, X(M_1), Y(M_2)) \mid \mathbf{u} : M \rightarrow M_1 \cup M_2\} \\(X \multimap Y)(M) &= \{f : \prod_{M'} (X(M') \rightarrow Y(M \cup M'))\}\end{aligned}$$

The category $\widehat{\mathcal{T}[\mathbf{N}]}$

The model of linear logic that we saw concerns with propositions and sets of markings – what happens when we move to the level of types and processes?

We go from $\mathcal{T}[\mathbf{N}]$ to the presheaves $\widehat{\mathcal{T}[\mathbf{N}]} = \mathcal{T}[\mathbf{N}]^{op} \rightarrow \mathbf{Set}$. The monoidal structure is then lifted:

$$\begin{aligned}1(M) &= \{\mathbf{U} : M \rightarrow \emptyset\} \\(X \otimes Y)(M) &= \{(\mathbf{U}, X(M_1), Y(M_2)) \mid \mathbf{U} : M \rightarrow M_1 \cup M_2\} \\(X \multimap Y)(M) &= \{f : \prod_{M'} (X(M') \rightarrow Y(M \cup M'))\}\end{aligned}$$

i.e. $\widehat{\mathcal{T}[\mathbf{N}]}$ is a closed symmetric monoidal category (interpretation for $\otimes, \multimap$) which contains $\mathcal{T}[\mathbf{N}]$ as a full subcategory

- $yM = \mathbf{Hom}_{\mathcal{T}[\mathbf{N}]}(-, M)$ is a collection of markings reaching into M
- $y(M \cup M') \simeq yM \otimes yM'$

There are also products $\times$ and coproducts $+$ which we use to interpret $\&$ and $\oplus$.

$\widehat{\mathcal{T}}[\mathbf{N}]$ as a model for $\pi\mathbf{MILL}$

This gives us interpretation of $\pi\mathbf{MILL}$, and with the cartesian structure, of additives.

$$\llbracket \Gamma \vdash P :: z : A \rrbracket : \llbracket \Gamma \rrbracket \rightarrow \llbracket A \rrbracket$$

The interpretation is **sound**: If $P \longrightarrow Q$ then $\llbracket P \rrbracket = \llbracket Q \rrbracket$.

$\widehat{\mathcal{T}}[\mathbb{N}]$ as a model for πMILL

This gives us interpretation of πMILL , and with the cartesian structure, of additives.

$$\llbracket \Gamma \vdash P :: z : A \rrbracket : \llbracket \Gamma \rrbracket \rightarrow \llbracket A \rrbracket$$

The interpretation is **sound**: If $P \longrightarrow Q$ then $\llbracket P \rrbracket = \llbracket Q \rrbracket$.

This is a “generic” interpretation that we have in any symmetric closed monoidal category.
We are interested in exposing/talking about PN-related content of the model!

Atomic processes: typing

$$\frac{u : (\bigcup_i M_i) \rightarrow M' \in \mathcal{T}[N]}{x_1 : M_1, \dots, x_n : M_n \vdash \bar{z}.\text{tr}_{\vec{x}}(u) :: z : M'}$$

Atomic processes: typing

$$\frac{u : (\bigcup_i M_i) \rightarrow M' \in \mathcal{T}[N]}{x_1 : M_1, \dots, x_n : M_n \vdash \bar{z}.\text{tr}_{\bar{x}}(u) :: z : M'}$$

is interpreted as

$$y u : \llbracket M_1 \rrbracket \otimes \dots \otimes \llbracket M_n \rrbracket \rightarrow \llbracket M' \rrbracket,$$

equivalently

$$y u : y(\bigcup_i M_i) \rightarrow y M'$$

Atomic processes: reductions

$$\frac{\frac{k : L \rightarrow M \in \mathcal{T}[N]}{l : L \vdash \bar{w}.tr_l(k) :: w : M} \quad \frac{t : (M \cup \bigcup_i M_i) \rightarrow M' \in \mathcal{T}[N]}{w : M, x_1 : M_1, \dots x_n : M_n \vdash \bar{z}.tr_{w\vec{x}}(t) :: z : M'}}{l : L, x_1 : M_1, \dots x_n : M_n \vdash (\nu w).(\bar{w}.tr_l(k) \parallel \bar{z}.tr_{w\vec{x}}(t)) :: z : M'}$$

Atomic processes: reductions

$$\frac{\frac{k : L \rightarrow M \in \mathcal{T}[N]}{l : L \vdash \bar{w}.tr_l(k) :: w : M} \quad \frac{t : (M \cup \bigcup_i M_i) \rightarrow M' \in \mathcal{T}[N]}{w : M, x_1 : M_1, \dots x_n : M_n \vdash \bar{z}.tr_{w\vec{x}}(t) :: z : M'}}{l : L, x_1 : M_1, \dots x_n : M_n \vdash (\nu w).(\bar{w}.tr_l(k) \parallel \bar{z}.tr_{w\vec{x}}(t)) :: z : M'}$$

$\longrightarrow$

$$\frac{(k \otimes id_{\bigcup_i M_i}); t : (L \cup \bigcup_i M_i) \rightarrow M' \in \mathcal{T}[N]}{l : L, x_1 : M_1, \dots x_n : M_n \vdash \bar{z}.tr_{l\vec{x}}((k \otimes id_{\bigcup_i M_i}); t) :: z : M'}$$

Session-Typed Calculus over a Petri net

Summary: for a Petri net N we define a calculus $\pi\text{MILL}(N)$ by extending πMILL to include markings as atomic types and atomic processes

$$\frac{u : (\bigcup_i M_i) \rightarrow M' \in \mathcal{T}[N]}{x_1 : M_1, \dots, x_n : M_n \vdash \bar{z}.\text{tr}_{\bar{x}}(u) :: z : M'}$$

Session-Typed Calculus over a Petri net

Summary: for a Petri net N we define a calculus $\pi\text{MILL}(N)$ by extending πMILL to include markings as atomic types and atomic processes

$$\frac{u : (\bigcup_i M_i) \rightarrow M' \in \mathcal{T}[N]}{x_1 : M_1, \dots, x_n : M_n \vdash \bar{z}.\text{tr}_{\bar{x}}(u) :: z : M'}$$

This calculus is interpreted in $\widehat{\mathcal{T}[N]}$ and the interpretation is sound w.r.t reduction:
 $P \longrightarrow Q \implies \llbracket P \rrbracket = \llbracket Q \rrbracket$.

Adequacy?

What is a suitable **adequacy** result for $\pi\text{MILL}(\mathbf{N})$ and $\widehat{\mathcal{T}}[\mathbf{N}]$?

Adequacy?

What is a suitable **adequacy** result for $\pi\text{MILL}(\mathbb{N})$ and $\widehat{\mathcal{T}}[\mathbb{N}]$?

In λ -calculus an adequacy result concerns closed terms: if $\cdot \vdash t : \text{nat}$ then $t \longrightarrow^* \llbracket t \rrbracket \in \mathbb{N}$.

Adequacy?

What is a suitable **adequacy** result for $\pi\text{MILL}(\mathbb{N})$ and $\widehat{\mathcal{T}}[\mathbb{N}]$?

In λ -calculus an adequacy result concerns closed terms: if $\cdot \vdash t : \text{nat}$ then $t \longrightarrow^* \llbracket t \rrbracket \in \mathbb{N}$.

In our case we cannot say a lot about closed processes $\cdot \vdash P :: z : M'$.

Much better to talk in general about semi-closed $x : M \vdash P :: z : M'$.
Such processes correspond exactly to maps $M \rightarrow M'$ in $\mathcal{T}[\mathbb{N}]$.

Adequacy

Theorem

If $\llbracket x_1 : M_1, \dots, x_n : M_n \vdash P :: z : M' \rrbracket = yU$ for $U : (\bigcup M_i) \rightarrow M'$, then $P \longrightarrow^ \bar{z}.\text{tr}_{\vec{x}}(U)$.*

Adequacy

Theorem

If $\llbracket x_1 : M_1, \dots, x_n : M_n \vdash P :: z : M' \rrbracket = yU$ for $U : (\bigcup M_i) \rightarrow M'$, then $P \longrightarrow^ \bar{z}.\text{tr}_{\vec{x}}(U)$.*

Proof idea: using an **open logical relation** [Barthe et al. 2020] between syntax and semantics

$$\mathcal{R}^{\vec{x}:\vec{M}}[z:A] \subseteq (\vec{x}:\vec{M} \vdash \cdot :: z:A) \times (\llbracket \vec{M} \rrbracket \rightarrow \llbracket A \rrbracket)$$

Adequacy

Theorem

If $\llbracket x_1 : M_1, \dots, x_n : M_n \vdash P :: z : M' \rrbracket = yU$ for $U : (\bigcup M_i) \rightarrow M'$, then $P \longrightarrow^* \bar{z}.\text{tr}_{\vec{x}}(U)$.

Proof idea: using an **open logical relation** [Barthe et al. 2020] between syntax and semantics

$$\mathcal{R}^{\vec{x}:\vec{M}}[z : A] \subseteq (\vec{x} : \vec{M} \vdash \cdot :: z : A) \times (\llbracket \vec{M} \rrbracket \rightarrow \llbracket A \rrbracket)$$

$$\mathcal{R}^{\vec{x}:\vec{M}}[z : M] = \{(P, yU) \mid P \longrightarrow^* \bar{z}.\text{tr}_{\vec{x}}(U)\}$$

$$\begin{aligned} \mathcal{R}^{\vec{x}:\vec{M}}[z : A \otimes B] &= \{(P, f \otimes g) \mid \forall Q, R, P \xrightarrow{\bar{z}[(\nu y).P]} R \text{ and } \vec{x} = \vec{x}'\vec{x}'' \\ &\implies (P, f) \in \mathcal{R}^{\vec{x}':\vec{M}'}[y : A] \text{ and } (Q, g) \in \mathcal{R}^{\vec{x}'':\vec{M}''}[z : B]\} \end{aligned}$$

Adequacy

Theorem

If $\llbracket x_1 : M_1, \dots, x_n : M_n \vdash P :: z : M' \rrbracket = yU$ for $U : (\bigcup M_i) \rightarrow M'$, then $P \longrightarrow^ \bar{z}.tr_{\vec{x}}(U)$.*

For $\Delta \vdash P :: z : A$ and $p : \llbracket \Delta \rrbracket \rightarrow \llbracket A \rrbracket$,

$$\begin{aligned} \Delta \models P \sim p :: z : A &\iff \forall(\vec{x} : \vec{M}), (\vec{y} : \vec{M}'), \Delta', (\Delta = \vec{x} : \vec{M}, \Delta') \implies \forall(\mathcal{C}[\cdot], d) \in \mathcal{R}^{\vec{y}:\vec{M}'}[\Delta'], \\ &\quad (\mathcal{C}[P], p \circ (\text{id} \otimes d)) \in \mathcal{R}^{\vec{x}:\vec{M}, \vec{y}:\vec{M}'}[z : A] \end{aligned}$$

Adequacy

Theorem

If $\llbracket x_1 : M_1, \dots, x_n : M_n \vdash P :: z : M' \rrbracket = yU$ for $U : (\bigcup M_i) \rightarrow M'$, then $P \longrightarrow^ \bar{z}.tr_{\vec{x}}(U)$.*

For $\Delta \vdash P :: z : A$ and $p : \llbracket \Delta \rrbracket \rightarrow \llbracket A \rrbracket$,

$$\begin{aligned} \Delta \vdash P \sim p :: z : A &\iff \forall(\vec{x} : \vec{M}), (\vec{y} : \vec{M}'), \Delta', (\Delta = \vec{x} : \vec{M}, \Delta') \implies \forall(\mathcal{C}[\cdot], d) \in \mathcal{R}^{\vec{y}:\vec{M}'}[\Delta'], \\ &\quad (\mathcal{C}[P], p \circ (\text{id} \otimes d)) \in \mathcal{R}^{\vec{x}:\vec{M}, \vec{y}:\vec{M}'}[z : A] \end{aligned}$$

By induction, show $\Delta \vdash P \sim \llbracket p \rrbracket :: z : A$ for all well-typed P .

Future (current?) work

- Flesh out the open logical relations proof

Future (current?) work

- Flesh out the open logical relations proof
- Figure out the story behind !
- Completeness and characterization of denotational equivalence

Future (current?) work

- Flesh out the open logical relations proof
- Figure out the story behind !
- Completeness and characterization of denotational equivalence
- Contextual equivalence: not enough observations

Any fully closed process $\cdot \vdash P :: z : 1$ has no meaningful barbs



Future (current?) work

- Flesh out the open logical relations proof
- Figure out the story behind !
- Completeness and characterization of denotational equivalence
- Contextual equivalence: not enough observations

Any fully closed process $\cdot \vdash P :: z : 1$ has no meaningful barbs

- We do not actually need the symmetry part – how much can we do with non-symmetric monoidal categories? E.g. starting with an arbitrary M we have a language to talk about $\widehat{M}$.

Thank you

Thank you for listening!

References I



Barthe, Gilles et al. (2020). “On the Versatility of Open Logical Relations – Continuity, Automatic Differentiation, and a Containment Theorem”. In: *Programming Languages and Systems – 29th European Symposium on Programming, ESOP 2020, Held as Part of the European Joint Conferences on Theory and Practice of Software, ETAPS 2020, Dublin, Ireland, April 25–30, 2020, Proceedings*. Ed. by Peter Müller. Vol. 12075. Lecture Notes in Computer Science. Springer, pp. 56–83. DOI: 10.1007/978-3-030-44914-8_3.



Caires, Luís, Frank Pfenning, and Bernardo Toninho (Mar. 2016). “Linear Logic Propositions as Session Types”. en. In: *Mathematical Structures in Computer Science* 26.3, pp. 367–423. ISSN: 0960-1295, 1469-8072. DOI: 10.1017/S0960129514000218.



Engberg, Uffe and Glynn Winskel (1994). “Linear Logic on Petri Nets”. In: *A Decade of Concurrency Reflections and Perspectives*. Ed. by J. W. de Bakker, W. -P. de Roever, and G. Rozenberg. Lecture Notes in Computer Science. Berlin, Heidelberg: Springer, pp. 176–229. ISBN: 978-3-540-48423-3. DOI: 10.1007/3-540-58043-3_20.



Grunter, Carl and Vijay Gehlot (Oct. 1989). *Nets as Tensor Theories*. Tech. rep. MS-CIS-89-68. University of Pennsylvania, p. 22.

References II



Meseguer, José and Ugo Montanari (Oct. 1990). “Petri Nets Are Monoids”. In: *Information and Computation* 88.2, pp. 105–155. ISSN: 0890-5401. DOI: 10.1016/0890-5401(90)90013-8. (Visited on 03/09/2024).